On the capabilities of large reasoning models for cardinal direction reasoning

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Abstract

Spatial reasoning has been an established area of research since around 1990 and a lot of interesting related work has been done by researchers in Geography, AI and Databases. With the arrival of large language models, there has also been research on evaluating their spatial reasoning capabilities. In this paper, we contribute to this emerging research area and evaluate 5 large reasoning models (GPT-5.4, gemini-3-flash-preview, phi-4, llama-4-maverick and magistral-small) on the task of computing the composition table of a well-known cardinal direction calculus.

Keywords

spatial reasoning, cardinal directions, composition table, large language models, large reasoning models

1. Introduction

In this paper, we evaluate the capabilities of five large reasoning models on the task of computing the composition table for the Cardinal Direction Calculus (CDC) defined in [1, 2, 3]. *Large Reasoning Models (LRMs)* [4] represent a recent paradigm shift in large language models, exhibiting strong capabilities in code generation, formal logic, and arithmetic deduction (e.g., OpenAI's o1 and Google DeepMind's Gemini 2.5). Building upon our previous work [5], we benchmark five state-of-the-art commercial and open-weight models not previously assessed: GPT-5.4, Gemini, Phi-4, Llama-4, and Magistral-Small.

We design three complementary experiments to evaluate these models under varying degrees of contextual knowledge:

- *Complete Context:* Models are prompted with the formal, complete definition of CDC.
- *Anonymized Context:* Following the protocol of [6], models receive the axiomatic definitions of CDC, but the relation names are masked to decouple spatial reasoning from semantic label associations.
- *Zero Context (Parametric Knowledge)*: Models are prompted with no formal definitions, evaluating their ability to solve the task relying solely on latent spatial knowledge acquired during pre-training.

The rest of the paper is organized as follows. Section 2 discusses related work. Section 3 presents our experiments. Section 4 concludes the paper. Finally, Appendix A presents the prompts used in our experiments.

2. Related Work

Spatial reasoning is usually formalized using qualitative calculi. A *qualitative calculus* is a formal system consisting of a set of, usually binary, relation symbols, syntactic rules for combining them and inference rules for reasoning with them. The simplest relations in a calculus are called *base* relations

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and every other relation is a disjunction of them. Base relations are typically defined so that they are *jointly-exhaustive* (i.e., they cover the whole space of possibilities of how two objects in the calculus can be related) and *pairwise-disjoint* (i.e., one and only one relation can hold between two objects in the calculus). This property of relations is usually written as *JEPD*.

For every calculus, it is common to define a *composition table* which contains the results of composing every base relation with every other base relation. If the relation R_1 holds between objects a and b and relation R_2 holds between objects b and c, the *composition* of R_1 and R_2, denoted by $R_1 \circ R_2$, is the relation between a and c which is implied by the conjunction of R_1 and R_2. The composition of two base relations is in general a disjunction of base relations. In this paper we deal with the composition of base cardinal direction relations.

As it is also pointed out in [7], the bibliography contains several calculi that are called "cardinal direction calculus". As an example, one can see the paper [8] where Frank studied two cardinal directions calculi based on points in 2-dimensional space, both using the same set of 9 JEPD directions ($N, S, E, W, NE, NW, SE, SW$ plus the identity EQ). The first calculus is formalized using eight equally-sized cone-shaped areas formed by four axes, while the second one using projections based on two orthogonal axes forming four equally-sized areas. The contributions of Frank's paper include the formal definition of the calculi and their composition tables, as well as the presentation of some ideas regarding applications and extensions with distance relations. In later work, Renz and Mitra [9] presented the general framework STAR which generalizes Frank's calculi.

[10] is another paper where Ligozat also studies cardinal directions among point objects in 2-dimensional space using the same set of 9 projection-based relations studied by Frank [8]. The main contributions of Ligozat's paper include introducing constraint networks as a way to encode sets of relations in the calculus, and studying the computational complexity of consistency-checking in these constraint networks.

The present paper uses the CDC calculus defined in the papers [1, 2, 3]. CDC is based on 9 JEPD relations: the cardinal directions North (N), South (S), East (E) and West (W), the intercardinal directions Northeast (NE), Southeast (SE), Northwest (NW) and Southwest (SW), and the relation Bounding Box (B) which captures that the primary object falls inside the bounding box of the reference object. The 9 relations are defined for regions that are homeomorphic to the closed unit disk in $\mathbb{R}^2$ and regions that are formed by finite unions of such regions. The formal definitions of these relations are given in Section 3.1 of [3] so we do not repeat them here. We only show two examples (see Figure 1). In the left part of the figure, object a (the primary object) is to the south of object b (the reference object). This is written formally as $a\ S\ b$ in [3]. In the right part of the figure, a part of object a is northeast of object b and another part of it is east of object b (written formally as $a\ S : SE\ b$). [3] calls the former kind of relations *single-tile relations* to distinguish them from *multi-tile relations* (the latter kind). We only deal with single-tile relations in this paper.

With the revolution of LLMs underway, some spatial reasoning researchers turned their attention to the following questions: "Can LLMs be used for spatial reasoning? Can they match the accuracy of existing symbolic reasoners?". The first paper to tackle these questions was [11] where Tony Cohn investigated the capabilities of ChatGPT-4 for topological reasoning in RCC-8 [12]. This paper was followed by [6] where Cohn and Blackwell investigated whether six LLMs (used through their APIs) can reason about mereotopological relations in RCC-8. They concentrated on three problems: getting an LLM to reconstruct the composition table of RCC-8, evaluating the alignment of LLM answers to human composition preferences, and reconstructing the conceptual neighbourhood of RCC-8 relations. [6] show that LLMs do not perform well on the above tasks; this is not surprising given that, e.g., for the computing the composition table, even humans might have difficulty doing it.

Cohn and Blackwell also investigated whether LLMs can reason about cardinal directions following a different approach [13]. Their methodology was to create two question and answer datasets which they call *small* and *large*. To create small, they have used ChatGPT to co-create 100 simple questions where the answer is a cardinal direction (North, South, East and West). Two example questions are: *"You are watching the sun set. Which direction are you facing?"* and *"If the South Pole is behind you, which direction are you facing?"*. The second dataset is generated from a set of templates and it is meant to

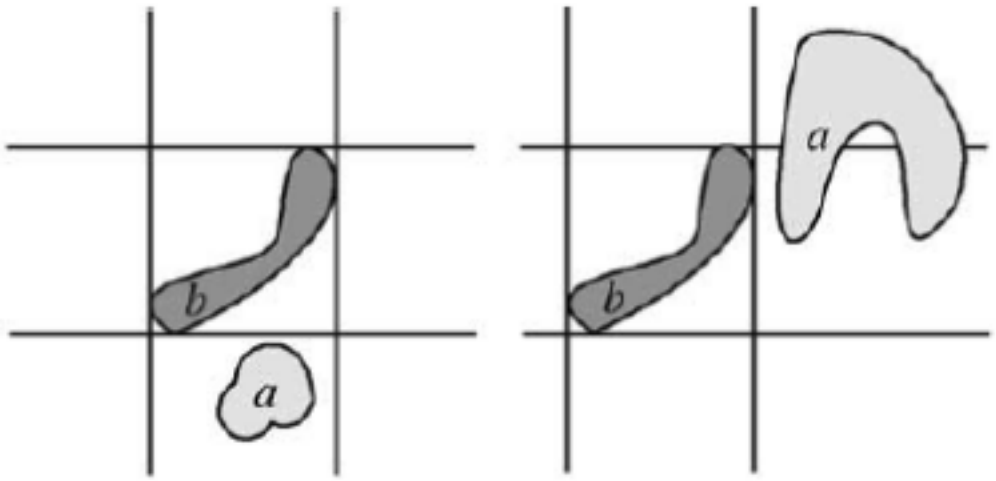


Figure 1: Two examples from the calculus of [3]

test comprehensively an LLM's ability to determine the correct cardinal direction given a particular scenario. An example question template from this dataset is *"You are walking [south] along the [east] shore of a lake and then turn around to head back in the direction you came from, in which direction is the lake?"*. Even with a temperature setting of 0, the experiments of [13] demonstrate that although LLMs are able to perform well in the small (simpler) dataset, in the second more complex dataset, no LLM is able to reliably determine the correct cardinal direction. Recently, Gyftou et al. [14] translated the large dataset into Greek and French and investigated the capabilities of ten multilingual LLMs on the translated datasets. GPT-5.2 is the model that shows the best performance in the Greek and French datasets (0.95 ± 0.01 and 0.94 ± 0.02 accuracy respectively) surpassing its performance in the English dataset (0.82 ± 0.01 accuracy).

In previous work, we evaluated the capabilities of 7 commercial LRMs (the OpenAI models o1, o3, and o4-mini, deepseek-r1, grok-3-mini, claude-3.7-sonnet with extended thinking mode and gemini-2.5-flash) on the reasoning tasks of computing the composition table of RCC-8 and CDC [5]. We showed that the model o3 outperforms the other 6 models in most of the experiments and achieves mean Jaccard index 99.66% and 99.57% for the CDC and the RCC-8 tasks respectively when the prompts included a precise definition of the two calculi.

More recently, Cohn and Blackwell proposed the benchmarks QSTRBench and QSTRBenchExtended for studying the temporal and spatial reasoning capabilities of 32 LLMs ranging from frontier commercial models to smaller open weights models [7]. They considered three qualitative temporal calculi and six qualitative spatial calculi and studied three reasoning problems for them: computing the inverse of a relation, computing the conceptual neighborhood of a relation, and reconstructing the composition table of a calculus. Among the spatial calculi tested are the projection-based cardinal direction calculi for points originally defined by Frank [8] which we mentioned above and the simplest version of the STAR calculus [9]. [7] shows that all models considered perform better than guessing, but none can answer all questions correctly. Performance varies by calculus, with the Point Algebra [15] being the easiest (16 out of 33 models evaluated answer all relevant questions correctly), and RCC-22 [16] the most difficult (none of the LLMs evaluated answers all questions correctly).

Finally, the paper [17] investigates whether LLMs can understand the widely-used Well-Known-Text representation of geometries and whether spatial relations (e.g., topological) are preserved during spatial reasoning when the corresponding vector data are passed to LLMs. [17] experiments with GPT-3.5-turbo, GPT-4, and DeepSeek-R1-14B and reports their accuracy in the identification of spatial relations when using geometry embedding-based, prompt engineering-based, and everyday language-based approaches. In particular, the GPT-based reasoners studied can understand inverse topological spatial relations.

In addition, GPT-4 can translate certain vernacular descriptions about places into formal topological relations, and adding the geometry-type or place-type context in prompts improves inference accuracy, although this varies by instance.

3. Experimental Setup

We perform our experiments on five models summarized in Table 1. The open-weight models phi-4-reasoning-plus, llama-4-maverick and magistral-small-2506 are accessed via OpenRouter. All models are evaluated using their default hyper-parameters. Henceforth, the models will be referred to by their abbreviations as listed in Table 1. The experiments examine the ability of the chosen LRMs to perform compositional reasoning. This is to reconstruct the composition table of CDC. Each experiment is conducted 3 times per model. The prompts used are presented in Appendix A.

To evaluate the answers the models give, we employ Jaccard Similarity since each composition may result in more than one possible relation. The Jaccard index is computed as the size of the intersection of the predicted set and the expected set of relations, divided by the number of relations in the union of the two sets. When only a single relation is predicted and the expected answer is also a single relation, the Jaccard index reduces to a binary 1 or 0 measure of accuracy. Since each experiment is conducted 3 times per model, we report in Table 2 the mean and standard deviation of the Jaccard Similarity averaged over all compositions and all 3 runs.

To visualize the performance across the full composition table, we present heatmaps (Figures 2, 6, 4) where each cell represents a composition and is shaded according to its mean Jaccard Index across the 3 runs ranging from 0 (dark purple) to 1 (bright yellow). For example, in the Figure 2a the composition $B \circ B$ has an average Jaccard Index of 1 and is in color yellow, while the composition $N \circ NE$ achieves 0.1667 and has a dark blue color. The relations in the cells of the figures are the ground truth relations. For the sake of readability, we use the letter U to denote the case where the ground truth result includes all relations.

To further analyze the behavior of the models, we decompose the prediction into binary classification counts for each relation r (Figures 3, 7, 5). We count correctly-predicted (TP) meaning the model correctly predicted the relation r as part of the result, correctly-not-predicted (TN) meaning the model correctly excluded relation r from the result, incorrectly-predicted (FP) meaning the model predicted the relation r as part of the result but it should not have, and incorrectly-not-predicted (FN) meaning the model did not predict the relation r but it should have. For example, in Figure 3a the model predicts the relation North as part of the result correctly 45 times (TP), it excludes it from its prediction correctly 154 times (TN), incorrectly includes it 14 times (FP) and incorrectly excludes it 30 times (FN) in all 3 runs of the experiment, summing to $45 + 154 + 14 + 30 = 243 = 81 \times 3$ total decisions across all $9 \times 9 = 81$ compositions and 3 runs.

Vendor	Model	Abbreviation	Released	API	Number of Parameters	Context
Microsoft	phi-4-reasoning-plus	phi-4-reasoning-plus	Apr 2025	OpenAI	14B	32K
Meta	llama-4-maverick-17B-128E-Instruct	llama-4-maverick	Apr 2025	OpenAI	400B(≈17B activated)	1M
Mistral	magistral-Small-2506	magistral-small	Jun 2025	OpenAI	24B	127K
OpenAI	GPT-5.4 Thinking	GPT-5.4	March 2026	OpenAI	Undisclosed	1M
Google	gemini-3-flash-preview	gemini-3-flash-preview	December 2025	Gemini API	Undisclosed	1M

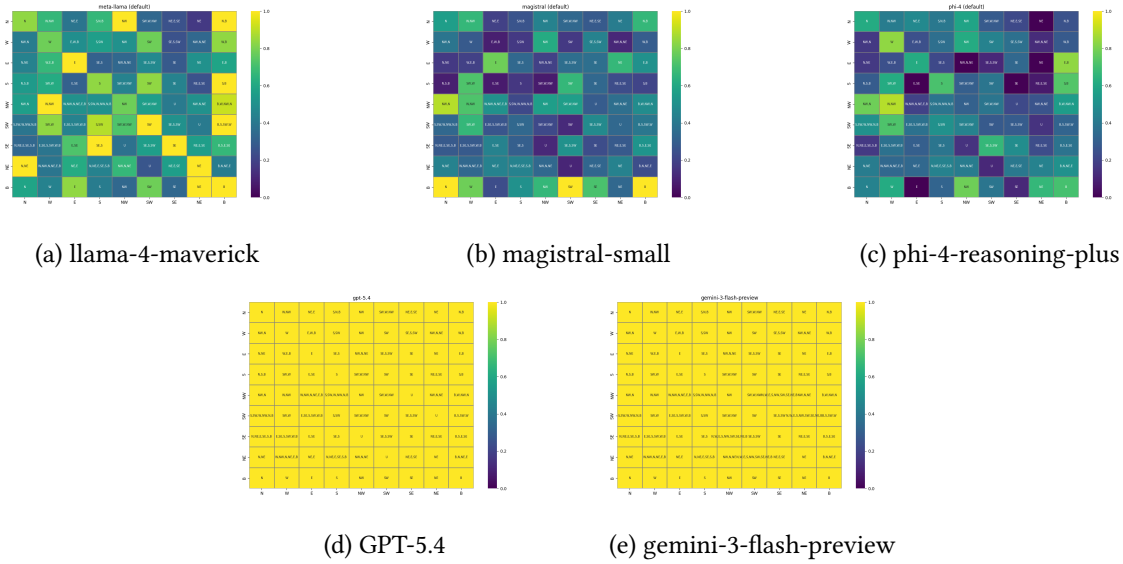
Table 1
LRMs tested. Context is the context window size in tokens. Models will be referred to by their abbreviation henceforth.

3.1. Open-Weight Models

First Experiment. We begin by evaluating the ability of the open-weight models tested to reconstruct the composition table of CDC when provided with the complete definition of the calculus and its

Model	With complete context	With anonymized context	With no context
phi-4-reasoning-plus	37.54 ± 2.92	-	14.58 ± 2.44
llama-4-maverick	60.04 ± 0.74	18.17 ± 1.01	20.08 ± 1.5
magistral-small	40.07 ± 2.5	-	16.23 ± 1.5
GPT-5.4	99.25 ± 1.3	31.73 ± 0.62	21.71 ± 0.79
gemini-3-flash-preview	100 ± 0	32.21 ± 0.24	20.28 ± 0.04

Table 2
Comparison of the mean and standard deviation of the Jaccard index achieved for the CDC experiment in three settings: with context included in the prompt, with anonymized context and with no context.



(a) llama-4-maverick (b) magistral-small (c) phi-4-reasoning-plus

(d) GPT-5.4 (e) gemini-3-flash-preview

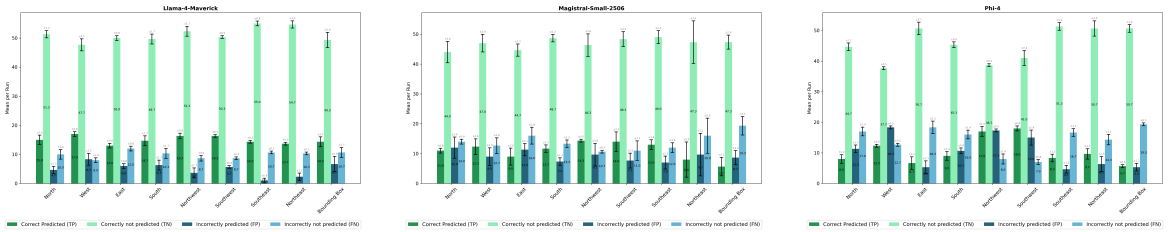
Figure 2: CDC composition table shaded by mean Jaccard index for the experiment with complete context.

relations as context. We present results for the open-weight models, phi-4-reasoning-plus, llama-4-maverick, and magistral-small using their default hyperparameters. As shown in Table 2, the models display varying degrees of success. Llama-4-maverick achieves the highest mean Jaccard index of 60.04%, followed by magistral-small at 40.07% and phi-4-reasoning-plus at 37.54%.

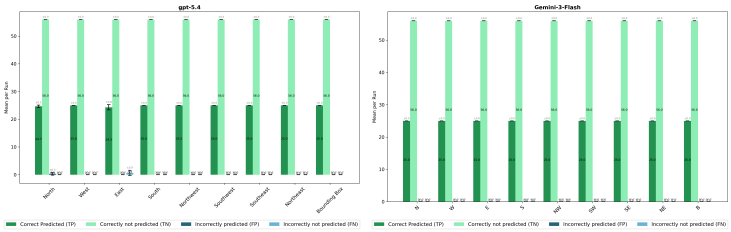
By inspecting Figure 2, llama-4-maverick shows many high-scoring cells, several in the diagonal $B \circ B$, $NE \circ NE$, $SE \circ SE$, $SW \circ SW$ and $E \circ E$, as well as some non-diagonal entries such as $NE \circ N$, $N \circ NW$, $N \circ B$. Magistral-small and phi-4-reasoning-plus produce largely dark heatmaps depicting the gap of approximately 20% of Jaccard index from the llama-4-maverick model. In particular, magistral-small achieves high scores only on $B \circ N$, $B \circ B$, $B \circ SW$ and phi-4-reasoning-plus on $NW \circ W$, $W \circ W$, $E \circ B$.

Figure 3 details each model's prediction behaviour across the nine spatial relations. For each relation, performance is decomposed into four classification outcomes: correctly-predicted (True Positives, TP), correctly-not-predicted (True Negatives, TN), incorrectly-predicted (False Positive, FP) and incorectly-not-predicted (False Negative, FN). Each bar reports the mean count per runs alongside its standard deviation (± std) to quantify variability across runs. This reporting format and metric layout are used consistently across all such figures.

All three models exhibit a pattern of high correctly-not-predicted (TN) counts relative to correctly-predicted (TP) counts, indicating that models are more confident in excluding a relation than in identifying one. For phi-4-reasoning-plus, the mean correctly-predicted (TP) values range from 5.7 (Bounding Box) to 18 (Southwest) against the mean correctly-not-predicted (TN) values of range 37.7 (West) and 51.3 (Southeast). Similarly, magistral-small, exhibits mean correctly-predicted (TP) values

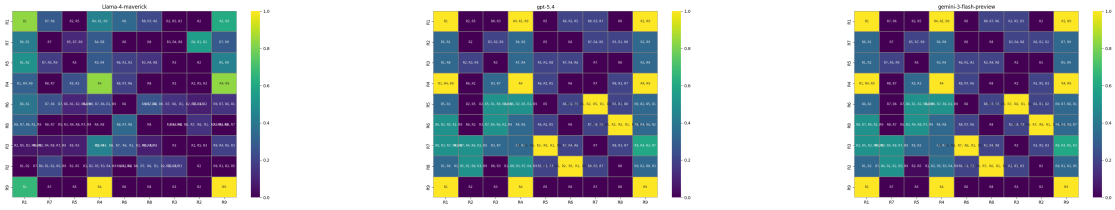


(a) llama-4-maverick (b) magistral-small (c) phi-4-reasoning-plus



(d) GPT-5.4 (e) gemini-3-flash-preview

Figure 3: Relation statistics for the CDC composition table computation for the experiment with complete context.



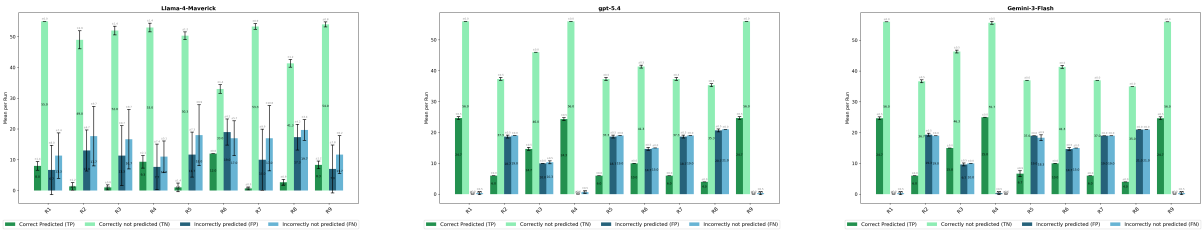
(a) llama-4-maverick (b) GPT-5.4 (c) gemini-3-flash-review

Figure 4: CDC composition table shaded by mean Jaccard index for the experiment with anonymized relations.

between 5.7 (Bounding Box) and 14.3 (Northwest) and correctly-not-predicted (TN) between 44 (North) and 49 (Southeast), while llama-4-maverick exhibits correctly-predicted (TP) values between 13 (East) and 17 (West) and correctly-not-predicted (TN) between 47.7 (West) and 55 (Southeast). Finally, the low standard deviation across runs ranging from $\pm$ 0.5 to $\pm$ 2.5 demonstrates that all three models exhibit stable and consistent prediction behaviors across iterations.

Second Experiment. Moving on, we evaluate the ability of the models to reconstruct the composition table of CDC when provided with the definition of relations, but now using anonymous labels R1 through R9 instead of the cardinal directions names. Due to the discontinuation of support on OpenRouter for the models phi4-reasoning-plus and magistral-small, this experiment is done only for llama-4-maverick. As shown in Table 2, the Jaccard index drops in this experiment for llama-4-maverick to 18.17%. Notably, the Jaccard index is close to that of the third experiment (see below), suggesting that good accuracy is achieved when a model is provided by both the definition of cardinal directions and the recognition of named cardinal directions.

Figure 4a illustrates the above behaviour. While the first experiment (Figure 2) yielded bright cells at multiple positions, Figure 4a shows only a small number of high scoring entries such as $R9 \circ R4$, $R9 \circ R9$, with the vast majority remaining dark. Figure 5 shows the same pattern identified in Figure 3. The correctly-not-predicted (TN) count, which ranges from 33 ($R6$) to 55 ($R1$), is substantially greater than the correctly-predicted count (TP), which ranges from 1 ($R7$) to 12 ($R6$). Interestingly, the standard deviation indicators reveal severe instability in the model's incorrect predictions across runs. While successful predictions correcly-predicted (TP) and correctly-not-predicted (TN) remain relatively

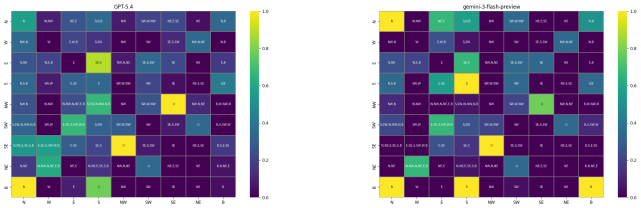


(a) llama-4-maverick (b) GPT-5.4 (c) gemini-3-flash-review

Figure 5: Relation statistics for the CDC composition table computation for the experiment with anonymized relations.



(a) llama-4-maverick (b) magistral-small (c) phi-4-reasoning-plus



(d) GPT-5.4 (e) gemini-3-flash-preview

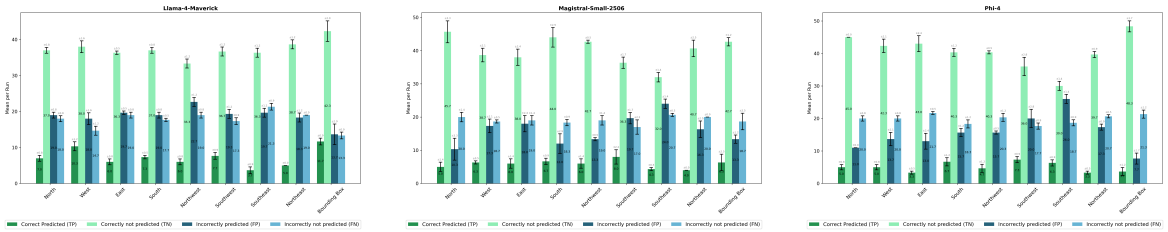
Figure 6: CDC composition table shaded by mean Jaccard index for the experiment without context.

stable ($\pm$ 0 to $\pm$ 2.9), both incorrectly-predicted (FP) and incorrectly-not-predicted (FN) mean values exhibit massive variance such as incorrectly-not-predicted standard deviation of $\pm$ 10.6 in R7 and incorrectly-predicted (FP) standard deviation of $\pm$ 8.0 in R1. This suggests substantial run-to-run volatility specifically when the model makes errors.

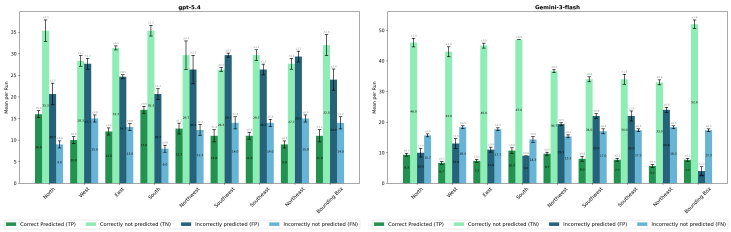
Third Experiment. Finally, we investigate whether the models can reconstruct the composition table without any contextual description of the cardinal direction relations and relying solely on their internal parametric knowledge. As shown in Table 2, all three models exhibit a sharp drop in performance compared to the first experiment. Specifically, phi-4-reasoning-plus, llama-4-maverick, and magistral-small achieve mean Jaccard similarities of 14.58%, 20.08%, and 16.23% respectively. This drop indicates that models lack sufficient parametric knowledge about cardinal direction composition to perform well without explicit guidance.

Figure 6 shows that all three models exhibit largely dark heatmaps reflecting the poor Jaccard indices they achieve. Llama-4-maverick exhibits high scores in cells $N \circ N$, $E \circ S$ and $S \circ S$ while magistral-small and phi-4-reasoning-plus only in one cell $N \circ N$ and $B \circ B$ respectively.

Figure 7 shows the same pattern we saw in the first experiment and second experiment i.e., of the correctly-not-predicted (TN) counts remaining high compared to correctly-predicted (TP). For example, llama-4-maverick records 5 correctly-predicted (TP) in the Northeast direction, while correctly-not-predicted (TN) reach 38.7 in the same direction. In addition, the incorrectly-not-predicted (FN) and incorrectly-predicted (FP) counts rise, reflecting poor performance since the models frequently fail to identify correct compositions. For instance, phi-4-reasoning-plus records 26 incorrectly-predicted (FP) relations in the Southeast direction and 21.7 incorrectly-not-predicted (FN) in East, and llama-4-maverick



(a) llama-4-maverick (b) magistral-small (c) phi-4-reasoning-plus

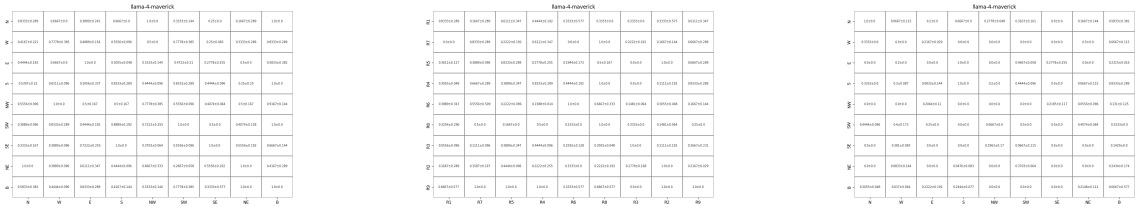


(d) GPT-5.4 (e) gemini-3-flash-preview

Figure 7: Relation statistics for the CDC composition table computation for the experiment without context.

reaches 22.7 incorrectly-predicted (FP) relations in Northwest and 21.3 incorrectly-not-predicted (FN) in Southeast. Similarly to the first experiment, the low standard deviation ranging from ± 0.5 to $\pm$ 2.9 indicates the models show stable and consistent behavior acroos the 3 runs.

Llama4-maverick model. The tables of Figure 8 show the mean and standard deviation of the Jaccard index for every composition computation in the case of the llama4-maverick model. Contrary to earlier figures that were heatmaps, now each cell shows explicitly the mean of the Jaccard index across all 3 repeats of the corresponding experiment and their standard deviation. Due to space limitations, we show this information only for the llama4-maverick model. The tables show that as the context decreases, the standard deviations in the three tables decrease as well. More specifically, the complete context experiment has higher standard deviations than the anonymized context experiment which also has higher standard deviations than the experiment with no context. This indicates that the more context we give the model about the CDC calculus (or any information we need it to handle in general), the more knowledge it has to handle, which leads to more randomness of the given answers and thus higher standard deviation.



(a) Complete Context (b) Anonymized Context (c) No Context

Figure 8: Mean and standard deviation of the Jaccard index for each composition computation for all experiments for the llama4-maverick model

3.2. Closed-Weight Models

First Experiment. We present results for the closed-weight models GPT-5.4 and gemini-3-flash-preview using their default hyperparameters. As shown in Figure 2, both models achieve near-perfect performance where gemini-3-flash-preview exhibits 100% and GPT-5.4 99.25% and their respective

heatmaps appear visually identical. For both models, the heatmap shows a highly consistent pattern, with all cells appearing at the maximum similarity level. The diagonal cells correctly preserve the expected relations, as well as the off-diagonal cells which contain the expected composed relation sets. However, some cells contain broader relation sets in GPT-5.4, especially in compositions involving NW, SW, SE, and NE explaining the small standard deviation of $\pm 1.3\%$.

Figure 3 shows that both closed-weight models perform almost perfectly across the nine spatial relations. For GPT-5.4, the correctly-predicted (TP) mean counts are very high and stable, since the counts mean ranges from 24.3 for East to 25 for West, for example and standard deviation is at most $\pm$ 0.5. Gemini-3-flash-preview shows an even more uniform pattern, with 25 mean TP and 56 mean TN for every spatial relation. It produces no FP or FN errors across all categories, suggesting that in this setting the model reconstructs the expected relation sets with complete consistency. Overall, both models demonstrate very strong performance, but gemini-3-flash-preview achieves a perfectly stable result across all relations, while GPT-5.4 shows only minimal errors in North and East.

Second Experiment. Moving on, we evaluate the ability of GPT-5.4 to reconstruct the composition table of cardinal directions provided with a textual description but using anonymous labels R1 through R9, for the cardinal directions names. As shown in Table 2, the Jaccard Index increases slightly in this case, compared to the second experiment. A likely explanation is that the anonymization in the second experiment removes the semantic cues that the model relies on. In this experiment the models may already have learned during pretraining that combinations such as $N \circ E$ result in relations involving NE, or that cardinal directions compose in certain intuitive ways. Therefore, it can use its general spatial knowledge and memorized/learned associations with cardinal directions to produce answer sets that overlap more with the expected CDC composition table cells.

Based on the results illustrated in Figure 4b, several cells reach a Jaccard index of 1.0, especially in cases involving identity-like or highly intuitive relations, such as $R1 \circ R1,\ R4 \circ R4,\ R9 \circ R9$, and some combinations involving $R1, R4, R7$, and $R8$. This indicates that the model can recover parts of the composition table even without being given the CDC definitions in the prompt. However, the heatmap also contains many low-similarity cells, especially around relations such as $R2, R3, R5, R6$, and $R8$, where the predicted sets only partially overlap with the expected sets or fail to overlap at all. This suggests that the model does not reconstruct the full algebra systematically, while instead, it seems to recover some relations through pretraining-based spatial intuition, while struggling with less direct or more abstract compositions.

Gemini-3-flash-preview is still able to recover parts of the composition structure, but its performance is uneven (Figure 4c). Several cells reach a Jaccard index of 1.0, while some off-diagonal cells also obtain very high similarity, suggesting that the model captures some repeated structural patterns among these anonymous relations, even though their original semantic meaning is hidden. Several cells contain long relation sets showing uncertainty: instead of predicting a precise composition result, the model sometimes includes many possible relations. Such broad predictions can increase the Jaccard index when part of the expected answer is included, but they also reveal that the model is not confidently identifying the exact composition relation.

Figure 5b further clarifies this behaviour across the nine relations where the strongest classes are North, South, and Bounding Box. GPT-5.4 correctly-predicted (TP) mean counts 24.7, 24.3 and 24.7 correspondingly, 56 correctly-not-predicted (TN) cases and 0 FP and 0 FN an these relations, indicating a stable behaviour for these relations. Based on these results, the third experiment suggests that GPT-5.4 has some pretrained knowledge of cardinal direction composition, particularly for the main cardinal directions and bounding-box relation, but this knowledge is uneven.

For gemini-3-flash-preview, the bar chart confirms the mixed behaviour observed in the heatmap (Figure 5c). The model performs accurately on R1, R4, and R9. These results show that gemini-3-flash-preview can reconstruct some anonymized relations almost perfectly, especially the relations that appear to have a stable or identity-like role in the composition table. However, the performance drops considerably for the remaining relations, suggesting that the model does not reconstruct the anonymized composition table in a fully systematic way, since it can recover some relations with very high accuracy, especially R1, R4, and R9, but it struggles with several other anonymized relations,

indicating both overprediction and omission of correct relations.

Third Experiment. Finally, we experiment with not giving the models any definition of CDC and its relations, but rather relying on the latent knowledge obtained by the model during its pretraining. As shown in Table 2, all models exhibit a sharp drop in performance compared to the first experiment, and the same occurs for gemini-3-flash-preview and GPT-5.4. This drop indicates that the models lack sufficient internal parametric knowledge about cardinal direction composition to perform without explicit guidance.

According to Figure 6, gemini-3-flash-preview performs best on cardinal relations, especially N, S, and B, which appear with very high Jaccard scores on the diagonal. However, its performance is weaker and more inconsistent for intercardinal relations such as NW, SW, SE, and NE. The model often produces broad sets of relations instead of a single precise relation, creating thus several off-diagonal cells with moderate or high Jaccard scores, because part of the predicted set overlaps with the expected relation; suggesting overgeneralization and confusion between neighboring or opposite cardinal directions. On the other hand, the results of GPT-5.4 show that the diagonal is not consistently strong: while some expected relations such as $N \circ N$ and $B \circ B$ achieve higher Jaccard similarity, other diagonal cells appear darker, indicating weaker overlap between the expected and predicted relation sets. There are also several bright off-diagonal cells, such as combinations involving South, Northwest, and Southeast, suggesting that the model sometimes assigns high similarity to relation pairs that should not be the main match, indicating that GPT-5.4 does not reconstruct the spatial composition structure perfectly in this setting; instead, it shows partial understanding, with noticeable confusion between intercardinal directions and broader relation sets.

Figure 7 presents the prediction behaviour of GPT-5.4 and gemini-3-flash-preview across the nine spatial relations. GPT-5.4 (Figure 7d) achieves higher correctly-predicted (TP) mean counts than gemini-3-flash-preview for every relation, with its strongest results observed for South (17), North (16.3), Northwest (12.7), and East (12). This indicates that GPT-5.4 is better at identifying relations that should be included in the expected answer set. However, the model also produces a high number of incorrectly-predicted (FP) cases, especially for Southwest (29.7), Northeast (29.3), West (27.7), Northwest (26.3), and Southeast (26.3). This suggests a tendency to overpredict, meaning that GPT-5.4 often includes additional relations that are not part of the correct output.

In contrast, gemini-3-flash-preview has lower TP mean counts overall, ranging from 5.7 for Northeast to 10.7 for South, showing that it identifies fewer expected relations correctly (Figure 7e). It achieves much higher correctly-not-predicted (TN) counts, particularly for Bounding Box (52), South (47), North (46), and East (45), indicating that gemini-3-flash-preview is more conservative and better at excluding relations that should not be predicted. However, this leads to many incorrectly-not-predicted (FN) cases, especially for West and Northeast (18.3), East (17.7), and Southeast and Bounding Box (17.3). Overall, GPT-5.4 shows stronger recall of the expected relations but tends to overgenerate, whereas gemini-3-flash-preview is more restrictive, producing fewer false positives but missing many correct relations.

Moreover, gemini-3-flash-preview exhibits stable and consistent prediction behavior shown by low standard deviation ranging from $\pm$ 0.5 to $\pm$1.4. In contrast, GPT-5.4 displays more variable behavior since standard deviation ranges from $\pm$ 0.5 to $\pm$ 2.4. Specifically, it exhibits low standard deviation on the correctly-predicted (TP) and incorrectly-not-predicted (FN) but high standard deviation on correctly-not-predicted (TN) and incorrectly-predicted (FP) which indicates that GPT-5.4 maintains stable behavior when identifying ground-truth relations, but highly variable behavior regarding how many incorrect relations it falsely predicts across different samples.

4. Summary and Future Work

We studied the capabilities of five large reasoning models (the closed-weight commercial models GPT-5.4 and gemini-3-flash-preview, and the open-weight models phi-4, llama-4-maverick and magistral-small) on the task of computing the composition table of the cardinal direction calculus CDC defined in [1, 2, 3].

We performed three experiments. The first experiment evaluated the accuracy of the models when prompted with the exact definition of the cardinal direction relations of CDC. The second experiment was like the first one, but, this time, relations were anonymized. The third experiment evaluated the accuracy of the models based only on the knowledge of cardinal direction relations stored on their parameters. The best results we got were in the first experiment; for the second and third experiment, there were significant drops in accuracy of the tested models. In the first experiment, the closed-weight commercial models gemini-3-flash-preview and GPT-5.4 achieved perfect and almost perfect accuracy respectively, while the open-weight models scored less satisfactorily, with llama-4-maverick scoring better than magistral-small and phi-4-reasoning-plus.

In future work we plan to experiment with other closed-weight and open-weight models and with more qualitative calculi, e.g., the full CDC calculus of [1, 2, 3] (therefore supporting multi-tile relations as well) and the calculus of [18] which combines topological and cardinal direction information.

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Declaration on Generative AI

We have experimented with the large reasoning models mentioned in the paper, but we have not used Generative AI in any way in the writing of the paper.

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A. Prompts used in the Experiments

In this appendix we present the prompts used in the experiments of Section 3. We start with the prompt used for the experiments with complete context. The prompt starts with some background information which essentially defines the CDC calculus and its base relations and then questions are posed to compute the composition table.

Background information (context): `Given a region a, the greatest lower bound or infimum of the projection of a on the x-axis (resp. on the y-axis) is denoted by infx(a) (resp. infy(a)). The least upper bound or the supremum of the projection of a on the x-axis (resp. on the y-axis) is denoted by supx(a) (resp. supy(a)). These bounds define the minimum bounding box of a region a, which is the box formed by the straight lines x=infx(a), x=supx(a), y=infy(a) and y=supy(a). Let us now consider regions that are homeomorphic to the closed unit disk` $\{(x, y) : x^2 + y^2 <= 1\}$. `The set of these regions will be denoted by REG. Regions in REG are closed, connected and have connected boundaries. A cardinal direction relation between regions in REG is one of the following relations: B (bounding box), S (South), SW (South West), W (West), NW (North West), N (North), NE (North East), E (East) and SE (South East). These relations are defined as follows: a B b if and only if infx(b) <= infx(a), supx(a) <= supx(b), infy(b) <= infy(a) and supy(a) <= supy(b). a S b`

if and only if supy(a) <= infy(b), infx(b) <= infx(a) and supx(a) <= supx(b). a SW b if and only if supx(a) <= infx(b) and supy(a) <= infy(b). a W b if and only if supx(a) <= infx(b), infy(b) <= infy(a) and supy(a) <= supy(b). a NW b if and only if supx(a) <= infx(b) and supy(b) <= infy(a). a N b if and only if supy(b) <= infy(a), infx(b) <= infx(a) and supx(a) <= supx(b). a NE b if and only if supx(b) <= infx(a) and supy(b) <= infy(a). a E b if and only if supx(b) <= infx(a), infy(b) <= infy(a) and supy(a) <= supy(b). a SE b if and only if supx(b) <= infx(a) and supy(a) <= infy(b). You are a helpful assistant. I will now give you a question regarding the cardinal direction relations I defined above. The possible answer can be one or more of N, NE, SE, S, E, NW, W, SW, B. No yapping.

Question. Let R_1 and R_2 be cardinal direction relations. If region x is R_1 of region y and region y is R_2 of region z, then which could the possible relations between region x and region z be?.

In the above question, variables R_1 and R_2 are relations from the set { N, NE, SE, S, E, NW, W, SW, B }.

Let us now give the prompt for the experiment with anonymized relations.

Backround information (context): Let a be a region on the Cartesian plane. We define functions f, g, h and g that apply to regions and return real values. The straight lines x=f(a), x=g(a), y=h(a) and y=i(a) define a box in the Cartesian plane. Let us now consider regions that are homeomorphic to the closed unit disk $\{(x,y) : x^2+y^2 <= 1\}$. The set of these regions will be denoted by REG. Regions in REG are closed, connected and have connected boundaries. A relation between regions in REG is one of the following: R1, R2, R3, R4, R5, R6, R7, R8 and R9. These relations are defined as follows (a and b are regions as above). a R9 b if and only if f(b) <= f(a), g(a) <= g(b), h(b) <= h(a) and i(a) <= i(b). a R4 b if and only if i(a) <= h(b), f(b) <= f(a) and g(a) <= g(b). a R6 b if and only if g(a) <= f(b) and i(a) <= h(b). a R2 b if and only if g(a) <= f(b), h(b) <= h(a) and i(a) <= i(b). a R5 b if and only if g(a) <= f(b) and i(b) <= h(a). a R1 b if and only if i(b) <= h(a), f(b) <= f(a) and g(a) <= g(b). a R8 b if and only if g(b) <= f(a) and i(b) <= h(a). a R3 b if and only if g(b) <= f(a), h(b) <= h(a) and i(a) <= i(b). a R7 b if and only if g(b) <= f(a) and i(a) <= h(b). You are a helpful assistant. I will now give you a question regarding cardinal direction relations. The possible answer can be one or more of R1, R2, R3, R4, R5, R6, R7, R8, R9. No yapping.

Question: Let S_1 and S_2 be relations. If region x is in relation S_1 with region y and region y is in relation S_2 with region z, then which could the possible relations between region x and region z be?.

In the above question, variables S_1 and S_2 are relations from the set {R1, R2, R3, R4, R5, R6, R7, R8, R9 }.

Let us now give the prompt for the experiment where we want to test whether LRMs can construct the composition table of CDC based only on their parametric knowledge.

Background information (context): You are a helpful assistant. I will give you a question regarding cardinal direction relations. The possible answer can be one or more of these: North, Northeast, Southeast, South, East, Northwest, West, Southwest, Bounding Box. No yapping.

Question: Let R_1 and R_2 be cardinal direction relations. If region x is R_1 of region y and region y is R_2 of region z, then which could the possible relations between region x and region z be?.

In the above question, variables R_1 and R_2 are relations from the set { North, Northeast, Southeast, South, East, Northwest, West, Southwest, Bounding Box }.